

Specifications

Volumetric Cementitious Mix-Pump System for Additive Construction

Background

The National Institute of Standards and Technology (NIST) Engineering Laboratory (EL) Materials and Structural Systems Division (MSSD) studies the durability and service life of infrastructure materials (polymer, concrete, engineered composites) for resilience infrastructure applications. One of the key research focuses is to evaluate mechanical properties of these materials either as a qualification of achieved strengths or the degradation of properties due to some external influence. As such the Additive Manufacturing with Cement-based Materials project seeks to further develop the measurement science tools and scientific knowledge base for performance-based standards for reinforced 3-D printed concrete structures

As part of MSSD, the Infrastructure Materials Group (IMG) currently owns a seven-axis robot mortar printer (RMP) outfitted with a new 2K print head (2 or more components), located in the Additive Construction Laboratory, of which is used to construct medium and large-scale specimens which will be tested in the newly equipped Structures Laboratory. As the project has developed alongside the rapidly improving industry, adaptation to multitude of systems being implemented into this field so that elements manufactured in the laboratory are representative of those being used throughout the industry during the standards and testing criteria are developed.

Purpose

The purpose of this acquisition is to procure a mix-pump system that can integrate into NIST's existing systems while permitting expanded print capabilities at an increased rate and accuracy. The system shall be volumetric - in that it permits continuous feeding with precision mixture rates.

Description of Required Products

*The Contractor shall provide **one** mix pump system that meets or exceeds all technical specifications identified below. All items must be new. Used or remanufactured equipment will not be considered for award. Experimental, prototype, or custom items will not be considered. The use of "gray market" components are not authorized for sale in the U.S. by the Contractor is not acceptable. All line items shall be shipped in the original manufacturer's packaging and include all original documentation and software, when applicable.*

Mix Pump System Minimum Specification/Requirements:

- 1) Connectivity:
 - a) Must be able to interface with current system for operation and sensor feedback.
 - i) 7-axis ABB robot (track system)
 - ii) XtreeE 2K print head (mixing nozzle)
 - iii) Single component nozzle
 - b) Adaptable to US electrical system
- 2) System specifications:
 - a) Stator with rotor (at least 5 spares)

- b) Dry material sensor
- c) Mortar temperature and pressure sensors (mixed material)
- d) Digital Incoming water pressure gauge
- e) Capable of delivering 20 L/min up to 40 m in distance
- f) Have connections for admixture introduction during mixing (plasticizers or retarders for example)

Maintenance Options

Not applicable

Warranty Requirements

The Contractor shall provide, at a minimum, a one-year warranty for the equipment after receipt of the equipment and shall be in accordance with terms stipulated in FAR 52.212-4. The warranty shall cover all parts, labor, and travel. The warranty shall commence upon acceptance of the system by the Government.

Delivery Requirements

Delivery shall be made F.O.B. Destination within 30 days after Contractor receipt of order to the National Institute of Standards and Technology at the address identified below.

National Institute of Standards and Technology (NIST)
100 Bureau Drive
Building 301 Shipping & Receiving
Gaithersburg, MD 20899-0001

FOB Destination means: The contractor shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery.

Alternatively, the government reserves the right to coordinate delivery of the subject product, in collaboration with the contractor, should it be determined to be in the best interest of the government to do so. Such a determination exclusively resides with the contracting officer and shall occur prior to the execution of any prospective contract.

Installation Requirements

Not applicable

Training Requirements

Not applicable

Inspection & Acceptance

In addition to the inspection and acceptance terms articulated in 52.212-4, the Government reserves the right to perform such performance tests and evaluations as defined below to verify specified system performance. Such tests and evaluations, if performed, shall be conducted within the

environment that the system is to be operated. The Contractor has the right to be present during the tests and evaluations, if performed, at the Contractor's expense.

Performance Tests:

NIST may choose at its discretion to forego this part of acceptance testing.

A visual inspection of the equipment will be performed by the NIST SME to identify surface defects or any form of indication that any equipment was damaged during transport to NIST. The Government shall have sole discretion to require repair or replacement of damaged and/or nonconforming supplies at no cost to the Government. The Government at any time prior to acceptance shall reject the equipment due to defects and/or nonconformance. The vendor is responsible for latent defects discovered any time after final inspection. However, the extent of its liability shall be prorated over the useful life of the equipment.

Ownership of the equipment shall transfer to NIST upon acceptance by the Government.

The Government will test, inspect, and accept or reject the equipment within 10 working days of the receipt of the equipment unless otherwise indicated above. The Government reserves the right to conduct quality assurance testing to confirm that a given instrument(s) meets the manufacturer's and/or the Government's performance specifications. It is anticipated that the equipment will meet all manufacturer's specifications and/or the Government's performance specifications identified in the most recent operations and maintenance manual for each piece of equipment and/or in this document.

Payment Schedule

Advance payment is not authorized. The Contractor must invoice in arrears according to the payment schedule. The Contractor shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. 100% after installation and acceptance by the SME of fully installed system, AND
2. After the successful completion of the testing requirements set forth in this document under section set forth in this document, AND
3. After successful demonstration by the instrumentation that it performs IAW the technical requirements set forth in this document AND
4. After receiving an invoice submitted properly, IAW the purchase order terms and conditions.

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer. Proposed payment schedules shall be submitted with vendor's response to the RFQ for consideration.